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Studierichting: Informatica
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$$f(x) = \frac{2x^2 - 4x + 1}{\sqrt{x}}$$

$$f'(x) = \frac{(4x-4)\sqrt{x} - (2x^2 - 4x + 1)\left(\frac{1}{2\sqrt{x}}\right)}{(\sqrt{x})^2}$$

$$= \frac{(4x-4)\sqrt{x} - \frac{2x^2 - 4x + 1}{2\sqrt{x}}}{x}$$

$$= \frac{\frac{2(4x-4)x - (2x^2 - 4x + 1)}{2\sqrt{x}}}{x}$$

$$= \frac{\frac{8x^2 - 8x - 2x^2 + 4x - 1}{2\sqrt{x}}}{x}$$

$$= \frac{\frac{6x^2 - 4x - 1}{2\sqrt{x}}}{x}$$

$$f'(x) = \frac{6x^2 - 4x - 1}{2x\sqrt{x}}$$

References